SkillCraft Technology – Task 3

Sudoku Game | Code Screenshot

```
def print_board(board):
    for row in board:
        print(" ".join(str(num) for num in row))

def is_valid(board, row, col, num):
    for x in range(9):
        if board[row][x] == num or board[x][col] == num:
            return False

    start_row = row - row % 3
    start_col = col - col % 3

    for i in range(3):
        for j in range(3):
            if board[start_row + i][start_col + j] == num:
                return False
    return True

def solve(board):
    for row in range(9):
        for col in range(9):
            if board[row][col] == 0:
                for num in range(1, 10):
                    if is_valid(board, row, col, num):
                        board[row][col] = num
                        if solve(board):
                            return True
                        board[row][col] = 0
                return False
    return True

board = [
    [5,3,0,0,7,0,0,0,0],
    [6,0,0,1,9,5,0,0,0],
    [0,9,8,0,0,0,0,6,0],
    [8,0,0,0,6,0,0,0,3],
```

SkillCraft Technology – Task 3

Sudoku Game | Successful Run Screenshot

```
PS C:\Users\aadikarthik> python sudoku.py

Sudoku Game
Initial Sudoku:
5 3 0 0 7 0 0 0 0
6 0 0 1 9 5 0 0 0
0 9 8 0 0 0 0 6 0
8 0 0 0 6 0 0 0 3
4 0 0 8 0 3 0 0 1
7 0 0 0 2 0 0 0 6
0 6 0 0 0 0 2 8 0
0 0 0 4 1 9 0 0 5
0 0 0 0 8 0 0 7 9

Solved Sudoku:
5 3 4 6 7 8 9 1 2
6 7 2 1 9 5 3 4 8
1 9 8 3 4 2 5 6 7
8 5 9 7 6 1 4 2 3
4 2 6 8 5 3 7 9 1
7 1 3 9 2 4 8 5 6
9 6 1 5 3 7 2 8 4
2 8 7 4 1 9 6 3 5
3 4 5 2 8 6 1 7 9

PS C:\Users\aadikarthik>
```

Demonstrated: initial Sudoku puzzle and successfully solved 9×9 board